

§4.3

Addition Rule for Probability

The probability that A or B occurs

Today's Goal

- Identify mutually exclusive (disjoint) events
- Apply the addition rule for mutually exclusive events
- Apply the general addition rule for any two events
- Calculate probabilities involving “or” statements

What You Will Be Graded On

LT.7 Probability

I understand what the probability of an event represents and can apply basic rules of probability to calculate the probability of an event.

Drawing One Card

One card is drawn from a standard deck of 52 cards.

**Can the card be a seven
and a nine? Can it be
a seven and a club?**

Mutually Exclusive Events

Definition 4.3.1

Two events are *mutually exclusive* (or *disjoint*) if they cannot occur at the same time—in other words, the events have no outcomes in common. Equivalently, $P(A \text{ and } B) = 0$.

Sevens, Nines, and Clubs

The sevens



The nines



A seven and a nine: mutually exclusive. A single card cannot be both a seven and a nine.

The clubs



A seven and a club: not mutually exclusive. The seven of clubs is both a seven and a club.

Mutually Exclusive or Not?

Determine whether each pair of events is mutually exclusive:

(a) Rolling a die: Event A = rolling a 5, Event B = rolling an even number.

Mutually exclusive. Rolling a 5 (odd) and rolling an even number cannot happen at the same time.

(b) Selecting a day of the week: Event A = a weekend day, Event B = Wednesday.

Mutually exclusive. Wednesday is not a weekend day.

Mutually Exclusive or Not?

Determine whether each pair of events is mutually exclusive:

(c) Selecting a day of the year: Event A = a day in December, Event B = a Monday.

Not mutually exclusive. Some days in December are Mondays.

(d) Rolling a die: Event A = rolling a number less than 4, Event B = rolling an odd number.

Not mutually exclusive. Rolling a 1 or 3 satisfies both conditions.

The Addition Rule

Theorem 4.3.4 Addition Rule for Probability

For any two events A and B , the probability that at least one of them occurs is given by

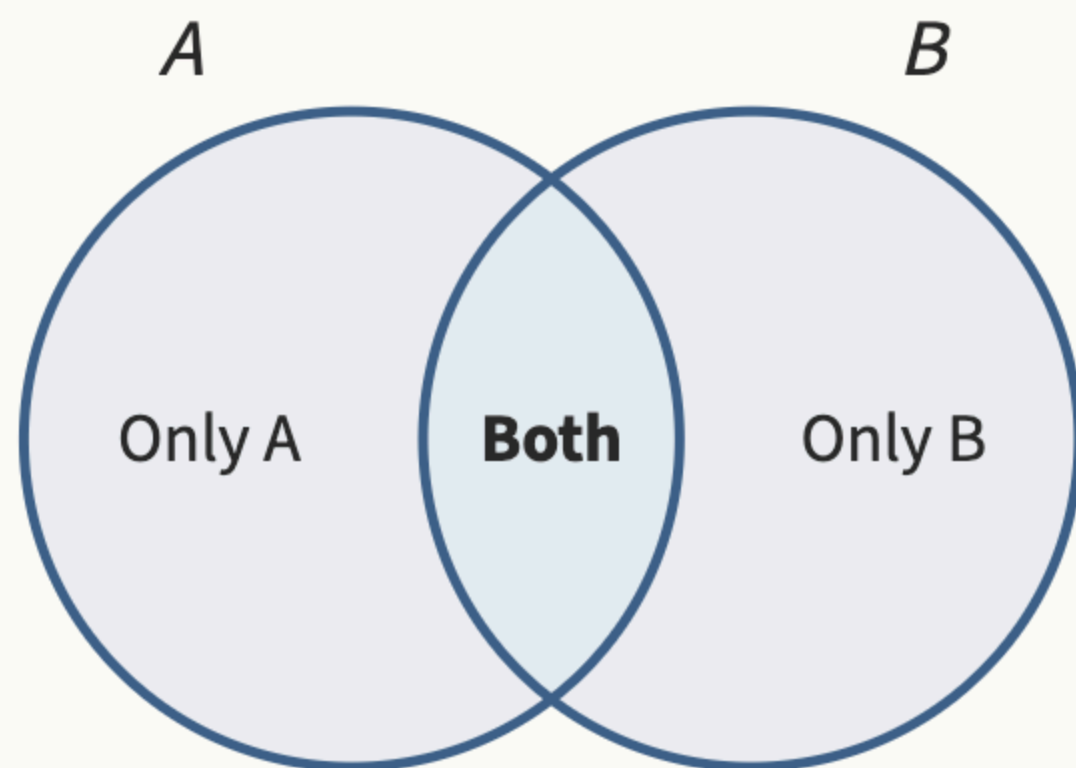
$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

When A and B are mutually exclusive, $P(A \text{ and } B) = 0$, and the rule simplifies to

$$P(A \text{ or } B) = P(A) + P(B).$$

The Addition Rule in a Venn Diagram

What goes wrong if we only add $P(A) + P(B)$?



The reason we subtract $P(A \text{ and } B)$ in the general rule is illustrated here: outcomes in both A and B would otherwise be counted twice.

An Ace or a Face Card

A card is randomly drawn from a standard deck of 52 cards. Find the probability of drawing an ace or a face card.

Let A = drawing an ace and B = drawing a face card. These events are mutually exclusive (a card cannot be both an ace and a face card).

- $P(A) = \frac{4}{52}$ (4 aces). $P(B) = \frac{12}{52}$ (12 face cards: jacks, queens, kings).

Using the addition rule for mutually exclusive events:

$$P(A \text{ or } B) = P(A) + P(B) = \frac{4}{52} + \frac{12}{52} = \frac{16}{52} = \frac{4}{13} \approx 0.308.$$

Addition Rule for More Than Two Events

The rule for mutually exclusive events extends to any number of mutually exclusive events:

$$P(A_1 \text{ or } A_2 \text{ or } \cdots \text{ or } A_k) = P(A_1) + P(A_2) + \cdots + P(A_k).$$

A fair twelve-sided die, numbered 1 through 12, is rolled. Find the probability of rolling a 1, 2, 3, or 4.

Rolling a 1, a 2, a 3, and a 4 are mutually exclusive events:

$$P(1 \text{ or } 2 \text{ or } 3 \text{ or } 4) = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}.$$

A Face Card or a Diamond

A card is randomly drawn from a standard deck of 52 cards. Find the probability of drawing a face card or a diamond.

Would the jack of diamonds be counted twice?

Let A = drawing a face card and B = drawing a diamond. These events are NOT mutually exclusive (the jack, queen, and king of diamonds belong to both).

- $P(A) = \frac{12}{52}$, $P(B) = \frac{13}{52}$, and $P(A \text{ and } B) = \frac{3}{52}$.

$$P(A \text{ or } B) = \frac{12}{52} + \frac{13}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26} \approx 0.423.$$

A Job or a Sport

In a group of 40 students, 24 have a part-time job, 18 play a campus sport, and 10 do both. If a student is selected at random, what is the probability that the student has a part-time job or plays a campus sport?

Let A = has a part-time job and B = plays a campus sport.

- $P(A) = \frac{24}{40}$, $P(B) = \frac{18}{40}$, and $P(A \text{ and } B) = \frac{10}{40}$.

$$P(A \text{ or } B) = \frac{24}{40} + \frac{18}{40} - \frac{10}{40} = \frac{32}{40} = \frac{4}{5} = 0.80.$$

Now You Try: Coffee or Tea

You Try 4.3.2. In a survey of 200 people, 120 like coffee, 80 like tea, and 50 like both. If a person is randomly selected from this group, find the probability that the person likes coffee or tea.

You Try 4.3.2: Checking Your Answer

200 people: 120 like coffee, 80 like tea, and 50 like both.

Coffee and tea are not mutually exclusive: 50 people like both.

$$P(\text{coffee or tea}) = \frac{120}{200} + \frac{80}{200} - \frac{50}{200} = \frac{150}{200} = \frac{3}{4} = 0.75$$

Using Tables to Find Probabilities

Contingency tables (two-way tables) are useful for organizing data and calculating probabilities involving two characteristics. This one shows 400 students.

	Walks	Drives	Takes the bus	Total
Lives on campus	90	10	20	120
Lives off campus	30	170	80	280
Total	120	180	100	400

Find the probability that a randomly selected student:

(a) Drives. $P(\text{drives}) = \frac{180}{400} = 0.45$

(b) Lives on campus and walks. $P(\text{on campus and walks}) = \frac{90}{400} = 0.225$

Using Tables to Find Probabilities

	Walks	Drives	Takes the bus	Total
Lives on campus	90	10	20	120
Lives off campus	30	170	80	280
Total	120	180	100	400

Find the probability that a randomly selected student:

(c) Lives off campus or takes the bus. $\frac{280}{400} + \frac{100}{400} - \frac{80}{400} = \frac{300}{400} = 0.75$

(d) Walks or takes the bus. $\frac{120}{400} + \frac{100}{400} = \frac{220}{400} = 0.55$

In (c), the 80 students in both are counted twice, so subtract them once. In (d), no student is in both columns.

Summary of Addition Rules

Situation	Formula
A and B are mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$
A and B are NOT mutually exclusive	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

What to Take With You

- The word “or” in probability is inclusive: it means A occurs, B occurs, or both occur.
- First ask: can both events happen on the same outcome?
- If they can, subtract $P(A \text{ and } B)$ once. If they cannot, just add.